

```
$ python3 day01.py
```

Python Day01

0000000000

```
>>> print("00000000000000")
```

```
00000000000000
```



layout: default hideInToc: true

```
$ cat agenda.md
```



1. Python Day01 — □□□□□□□□□□

2. □□□□

3. □□□□

4. □□

5. □□□□

6. □□□□□

7. □□□□□□□

8. Output

9. □□□□□□

10. f-string □□□□□

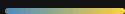
11. Input

12. □□□□

13. □□□□

layout: section transition: fade

PART 01



transition: slide-up



Python

(type)

10

int
□□

12.3

float
□□□

'hi'

str
□□

True

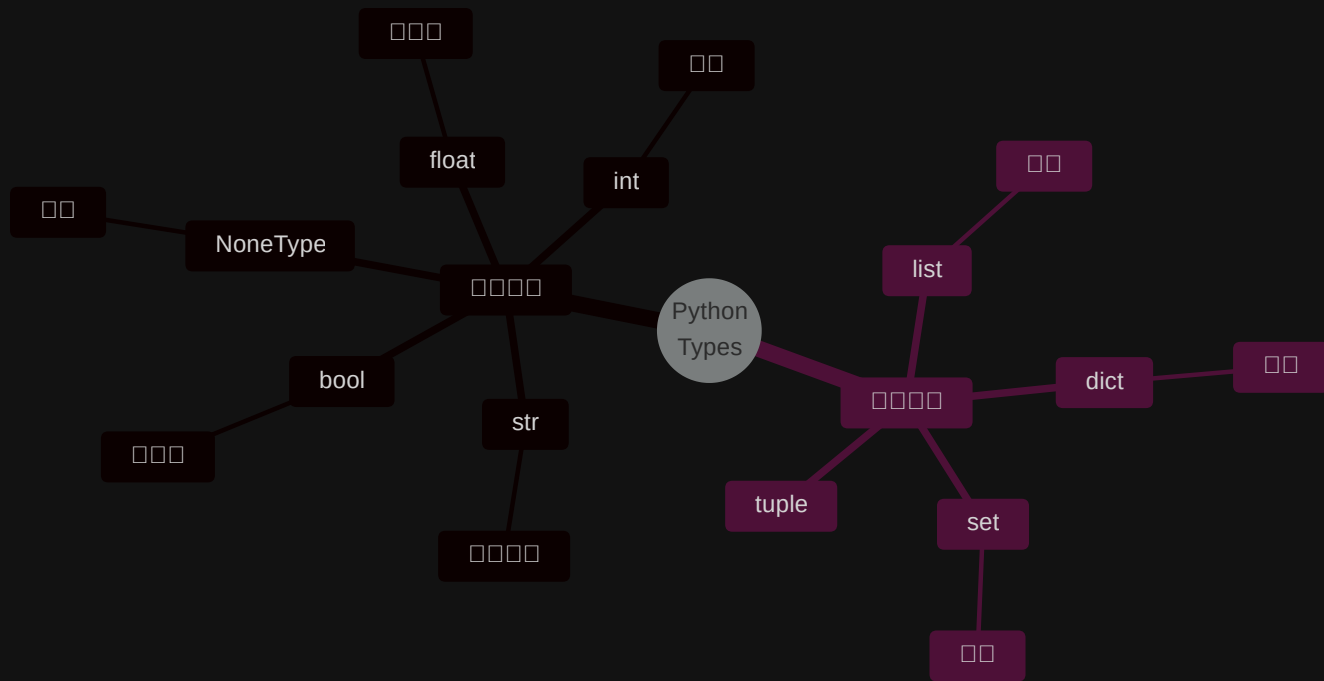
bool
□□

None

NoneType
□□

```
>>> type(10) # <class 'int'>
>>> type('hi') # <class 'str'>
```

transition: slide-up



transition: slide-up



000000

'123' ~ 00

123 ~ 00

'123' == 123 → False

'123'



123

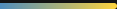


>>> '123' == 123

False # 0000000000

layout: section transition: fade

PART 02






[ML DSA](#)
[ML](#)
[DS](#)
[ML+DS](#)
[ML+DS+Python](#)
[ML+DS+Python+SQL](#)
[ML+DS+Python+SQL+Stats](#)
[ML+DS+Python+SQL+Stats+Math](#)
[ML+DS+Python+SQL+Stats+Math+System Design](#)
[ML+DS+Python+SQL+Stats+Math+System Design+Networking](#)
[ML+DS+Python+SQL+Stats+Math+System Design+Networking+OS](#)
[ML+DS+Python+SQL+Stats+Math+System Design+Networking+OS+Linux](#)
[ML+DS+Python+SQL+Stats+Math+System Design+Networking+OS+Linux+Cloud](#)
[ML+DS+Python+SQL+Stats+Math+System Design+Networking+OS+Linux+Cloud+DevOps](#)
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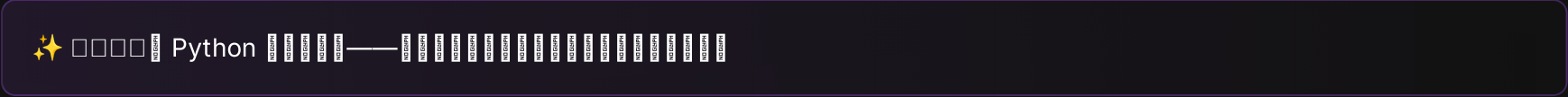


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if, for, while



layout: section transition: fade

PART 03

Output

```
❏ print() [] [] [] [] []
```

transition: slide-up



```
1 print('hi')
2 print('hi')      # 00000000 hi 000
```

00000

```
1 hi
2 hi
```



```
1 # 🤔 000 - 000
2 name = 'Lucas'
3 age = 18
4 print('Hello,', name, 'you are', age, 'years old')
```

```
1 # 😊 0 f-string - 0000
2 name = 'Lucas'
3 age = 18
4 print(f'Hello, {name}! You are {age} years old.')
```

```
1 # 🤖 f-string 0000000
2 name = 'Lucas'
```




print()



□□□□ □□□□ □□□□ □□□□

```
1 print('hi')
2 print('hi')
3 # :
4 # hi
5 # hi
```

🔑 end — '\n' · sep — ' ' □

f-string

f-string

```
1  # 0000
2  print(f'{{}}')
3
4  # 0000000000000000
5  pi = 3.14159265
6  print(f'π ≈ {pi:.2f}')    # π ≈ 3.14
7  print(f'π ≈ {pi:.4f}')    # π ≈ 3.1416
8
9  # 0000
10 print(f'{{42:05d}}')      # 00042
```

0000

```
{{:0}}
:.2f  - 0000
:05d  - 0000
```

 0000

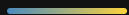
0000000000000000

```
1  pi = 3.14159265
2  print(f'π ≈ {pi:.2f}')
3  print(f'π ≈ {pi:.4f}')
4  print(f'{{42:05d}}')
```

layout: section transition: fade

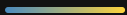
PART 04

Input



□□□□□□□□□□

transition: slide-up



layout: two-cols layoutClass: gap-8 transition: slide-up



 `input()`

```
1 n = input()
2 nn = input('0000000')
```

 `input()`
    `'123' != 123`   



□□□□□□□□□□□□□□□□

sys.stdin while + try □□□□

```
1  import sys
2
3  for line in sys.stdin:
4      line = line.strip()
5      print(line)
```

 sys.stdin

□□□
□□□□□□

 while+try

□□□□
□□□□□□

 □□□□

□□□
□□□□□□

transition: slide-up



 a.py 

```
1 a = input()
2 b = input()
3 print(a, b)
```

 in.txt 

```
1 Everything will get better
2 12345
```



```
1 python a.py < in.txt
```



```
1 Everything will get better 12345
```



□□□□□□□□

  in.txt □□□□□□□□□□□□□□□□

```
1 import sys
2
3 # UTF-8
4 sys.stdin.reconfigure(encoding='utf-8')
5
6 for line in sys.stdin:
7     line = line.strip()
8     print(line)
```

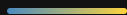
□□□□□□□□

```
1 python a.py < in.txt > out.txt
```

 UTF-8 Windows□□□□□□

layout: section transition: fade

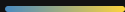
PART 05



transition: slide-up



□□	□□□	□□	□□□□
□□	<code>n + 2</code>	Addition	<code>n=10 → 12</code>
□□	<code>n - 2</code>	Subtraction	<code>n=10 → 8</code>
□□	<code>n * 2</code>	Multiplication	<code>n=10 → 20</code>
□□	<code>n / 2</code>	Division integer division float division float□	<code>n=10 → 5.0</code>
□□□□	<code>n // 2</code>	□□□□□□□□	<code>n=10 → 5</code>
□□	<code>n ** 6</code>	Power	<code>n=10 → 1000000</code>
□□□	<code>n % 6</code>	Modulo □□□□□□□□□□	<code>n=10 → 4</code>



```
1 n = 10
```

```
n + 2 → 12
```

```
n - 2 → 8
```

```
n * 2 → 20
```

```
n / 2 → 5.0 ← float
```

```
n // 2 → 5
```

```
n ** 6 → 1000000
```

```
n % 6 → 4 ← 10 ÷ 6 商
```

💡 /  float `//` `□□□□□□□□□□`

□ □ □ □ □ □ □

□ □ □ □ □ □ □ □ □ □

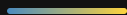
□ □ □ □ □ □ □ □ □ □

```
1  x = 10
2  x = x + 5    # x 15
3  x = x * 2    # x 30
4  print(x)
```

⚡ `x += 1` `10` `11` `12` `13` `14` `15` `16` `17` `18` `19` `20` `21` `22` `23` `24` `25` `26` `27` `28` `29` `30` `31` `32` `33` `34` `35` `36` `37` `38` `39` `40` `41` `42` `43` `44` `45` `46` `47` `48` `49` `50` `51` `52` `53` `54` `55` `56` `57` `58` `59` `60` `61` `62` `63` `64` `65` `66` `67` `68` `69` `70` `71` `72` `73` `74` `75` `76` `77` `78` `79` `80` `81` `82` `83` `84` `85` `86` `87` `88` `89` `90` `91` `92` `93` `94` `95` `96` `97` `98` `99` `100`

layout: section transition: fade

PART 06



layout: two-cols layoutClass: gap-8 transition: slide-up

□□□□□□□□

□□□□□□□□□□

- ✓ True □□□
- ✗ False □□□

□□□	□□
<	□□
<=	□□□□
>	□□
>=	□□□□
==	□□
!=	□□□

Logical Operators

Python Logical Operators

Operator	Description
<code>and</code>	Both operands must be true
<code>or</code>	At least one operand must be true
<code>not</code>	Reverse the logical state of the operand

```
1 age = 18
2 has_id = True
3
4 age >= 18 and has_id    # True
5 age < 18 or has_id     # True
6 not has_id              # False
```

🧠 `'and'` = Both operands must be true · `'or'` = At least one operand must be true · `'not'` = Reverse the logical state of the operand

Truthy / Falsy □□□□□□

Python True / False □□□□□□

```
1 bool(0)           # False
2 bool(1)           # True
3 bool("")          # False
4 bool("hi")        # True
5 bool(None)        # False
6 bool([])          # False
```

Falsy False

- 0
- " "
- None
- []
- {}

Truthy True

- 0
-
-

if □□□□

□□□□□□□□

```
1 age = 18
2
3 if age >= 18:
4     print("□□□□")
```

⚠ Python □□□□□□□□□□□□□□□□

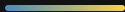
if / elif / else

□□□□□□

```
1  score = 85
2
3  if score >= 90:
4      print("A")
5  elif score >= 80:
6      print("B")
7  elif score >= 70:
8      print("C")
9  else:
10     print("F")
```



if score >= 90: print("A")
elif score >= 80: print("B")
elif score >= 70: print("C")
else: print("F")



```
1 username = input("000")
2 password = input("000")
3
4 if username == "admin" and password == "1234":
5     print("0000")
6 else:
7     print("0000")
```

🔑 000000input + 00 + and 0000

3



- Python □□□□□int / float / str / bool / None

- `input()` □□□□□□□□ □□

- `print()` □□□□□□□□ f-string □□□

- `/` □ `//` □ `%` □□□□□□□□

- □□□□□□□□ + □□□□□□□□

- `if` / `elif` / `else` □□□□

